FIITJEE

VISAKHAPATNAM CENTRE

NTSE - NOVEMBER, 2019 (stage-1 - AP)

MENTAL ABILITY TEST

QUESTION PAPER CODE: 109213

TIME : 2 hrs. **Max. Marks : 100**

INSTRUCTIONS

Read the following instructions carefully before you answer the questions. Answers are to be SHADED on a SEPARATE OMR Answer Sheet given, with HB Pencil. Read the **Instructions printed on the OMR Sheet carefully before answering the question.**

1. Please write your Center Code Number and Roll Number very clearly (only one digit in one block) on the OMR Sheet as given in your admission card. Please see that no block is left unfilled and even zeros appearing in the Center Code Number are correctly transferred to the appropriate blocks on the OMR Sheet as shown in the example given in the OMR Sheet. For all the subsequent purpose your Center Code Number and Roll Number shall remain the same as given on the Admission Card.
2. Paper-I (Mental Ability) consists of 100 questions (Q. Nos 1 to 100)
3. All questions carry one mark each.
4. Since all questions are compulsory do not try to read through the whole question paper before beginning to answer it.
5. Begin with the first question and keep trying one question after another till you finish all the questions
6. If you do not know the answer to any question, do not spend much time on it and pass on to next one. If time permits, you can come back to the questions which you have left in the first instance and try them again.
7. Since the time allotted to the question paper is very limited, you should make the best use of it by not spending too much time on any question.
8. A black page is provided for rough work at the end of question paper.
9. REMEMBER YOU HAVE TO SHADE ANSWERS ON A SEPARATE OMR SHEET PROVIDED.
10. Answer to each question is to be indicated by SHADING the circle having the number of the correct alternative in OMR Sheet from among the ones given for the corresponding question in the booklet.
11. Now turn to the next page and start answering the questions.
12. The OMR answer sheet consists of two copies, the ORIGINAL COPY and the CANDIDATE'S COPY. Do not separate or displace them. Do not darken the bubbles in two copies of OMR answer sheets separately. After the examination, you should hand over the OMR Sheet to the Invigilator of the room and can take away the candidate's copy of OMR answer sheet with them.

PAPER - I
MENTAL ABILITY TEST
(Q.Nos. 1 to 100)
Max. marks : 100

Note: SHADE the correct alternatives in the OMR Answer Sheet provided, from amongst the ones given against the corresponding question in the Question Booklet. For shading the circles, use HB Pencil.

Direction: In Question nos. 1 to 5:

Read the following and answer the answer the questions given below:

Ina united family, there are six members.

i) They are A, B, C, D, E and F.

ii) A and B are married couple.

iii) A is a male member.

iv) D is the only son of C, who is the brother of A.

v) E is the sister of D.

vi) B is the daughter-in-law of F, whose husband has died.

1. How is E related to C?

   1) Daughter 2) Sister 3) Mother 4) Mother-in-law

2. How many male members are there in the family?

   1) One 2) Two 3) Three 4) Four

3. How is F related to C?

   1) Sister-in-law 2) Mother 3) Sister 4) Mother-in-law

4. Who is C to B?

   1) Son 2) Brother-in-law 3) Nephew 4) Brother

5. How is F related to A?

   1) Sister 2) Sister-in-law 3) Mother 4) Mother-in-law

Direction: In questions nos. 6 to 15:

In the number series given below, one number is missing. Each series is followed by four alternatives (1), (2), (3) and (4). One of them is the right answer.

Identify and indicate it as per the "Instructions".

6. 12, 15, 27, 42, 69, 111, _____

   1) 164 2) 174 3) 160 4) 180

7. 0, 9, 21, 36, 54, _____

   1) 69 2) 77 3) 75 4) 144

8. 26, 4, 20, 10, 14, 16, 8, 22, 2, 28, _____

   1) 34 2) – 4 3) 6 4) – 6

9. 789, 678, 567, 456, 345, _____

   1) 254 2) 244 3) 234 4) 111

10. 17, 37, 77, 157, _____

    1) 217 2) 347 3) 417 4) 317

11. 39, 56, 73, 90, 107, 124 _____

    1) 147 2) 161 3) 137 4) 141

12. 6, 24, 60, 120, _____

1) 240 2) 224 3) 180 4) 210

13. 4, 32, 8, 64, 16, 128, _____

1) 32 2) 48 3) 156 4) 64

14. 14, 11, 13, 10, 12, 9, _____

1) 6 2) 15 3) – 3 4) 11

15. 0, 6, 24, 60, 120, _____

1)n 206 2) 216 3) 210 4) 196

Direction: In question nos. 16 to 20:

Some letters are given in Column I and some digits are given in Column II. Each digit of Column II represents any letter of Column I. Study the columns and write the alternative letter after choosing the correct alternative against the corresponding question.

Column – I	Column - II
DMBQZ	67024
ANYQD	84917
MBTYC	58603
TQCNM	54316
BDZAT	72509
ZQYAB	48902
QNTYM	41586
YBTCZ	80532
ZNMAC	29631
MTQYZ	46528

16. The product of the codes D and N is _____

1) 21 2) 7 3) 5 4) 8

17. The code for A is _____

1) 0 2) 9 3) 3 4) 8

18. The code for T is _____

1) 8 2) 5 3) 6 4) 3

19. The code for M is _____

1) 6 2) 5 3) 4 4) 2

20. The sum of the codes T and Q is _____

1) 9 2) 10 3) 11 4) 12

Direction: In Questions nos. 21 to 30:

In each of the following questions, a letter series is given, in which some letters are missing. The missing letters are given in the proper sequence as one of the alternatives. Find the correct alternative.

21. A_BAB_AB_ABB

1) BBB 2) BBA 3) AAA 4) ABB

22. PQR_ _RSPRS_ _SPQ

1) QRSPQ 2) RPQSQ 3) SQPQR 4) SPQPQ

23. _ _ABA_ _BA_AB

1) ABBAB 2) BAABB 3) BBABA 4) ABBBA

24. AC_CAB_BACA_ABA_ACAC

1) ACBC 2) AACB 3) BABB 4) BCBB

25. ABB_ _AB_B_BBA_A

1) BABBA 2) BBBAB 3) BBABB 4) ABAAB

26. _BAA_BBB_AB_

1) BBAB 2) ABAB 3) BAAB 4) AABB

27. W_V_ _XV_W_VW_XV

1) XWVXWV 2) XWWWXW 3) XVWXWX 4) VWXVWX

28. UV_V_U_WV_UV_VUU_WVU

1) WUVUWV 2) WVVUWV 3) UVWVUW 4) VUWVUW

29. _op_mo_n_ _pnmop_

1) mpnmop 2) mnpmon 3) mnpomn 4) mnompn

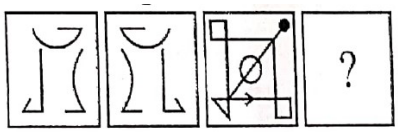
30. C_BBA_CAB_AC_AB_AC

1) BABCC 2) ACBCB 3) BCACB 4) ABCBC

Direction: In questions nos. 31 to 40:

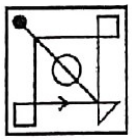
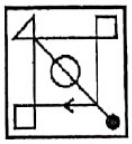
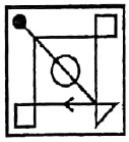
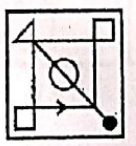
The following questions consist of two sets of figures A, B, C and D constitute the answer set. A definite relationship exists between figures A and B. You are required to establish a similar relationship between figures C and D by choosing a suitable figure D from the answer set.

31. Problem Figures:



(A) (B) (C) (D)

Answer Figures:

1)  2)  3)  4) 

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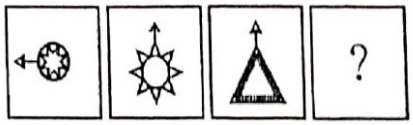
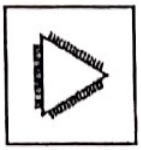
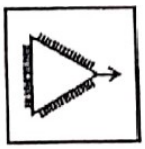


(A) (B) (C) (D)

Answer Figures:

1)  2) 3) 4) 

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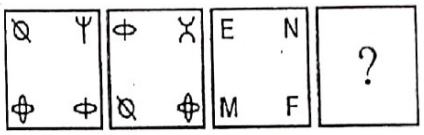

(A) (B) (C) (D)

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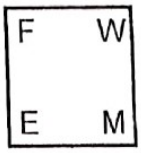
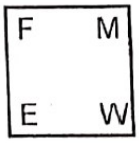
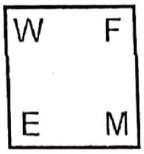
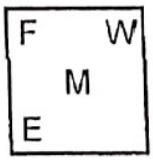
1) 

2) 

3) 

4) 

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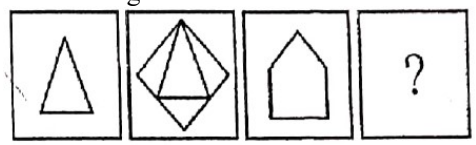
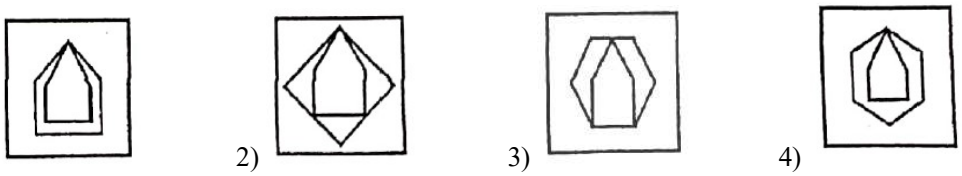


(A) (B) (C) (D)

Answer Figures:



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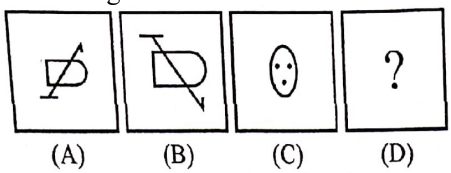

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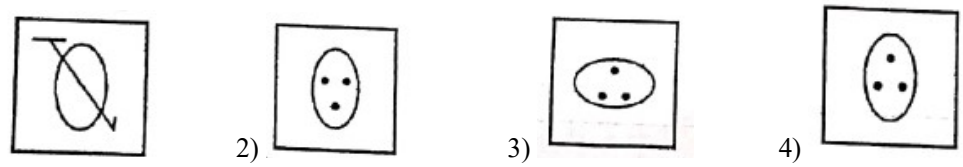

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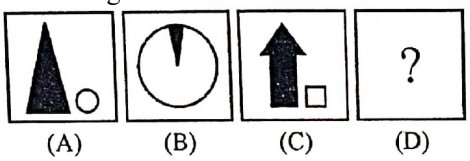

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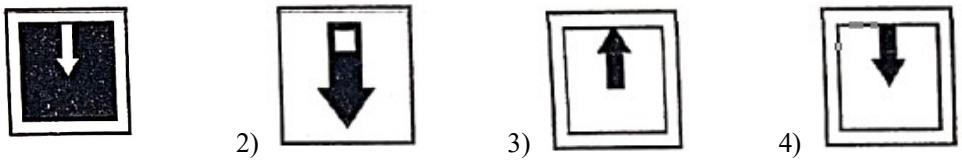

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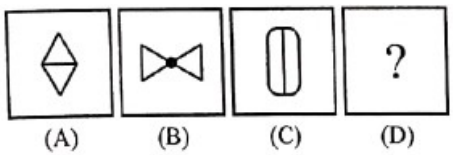

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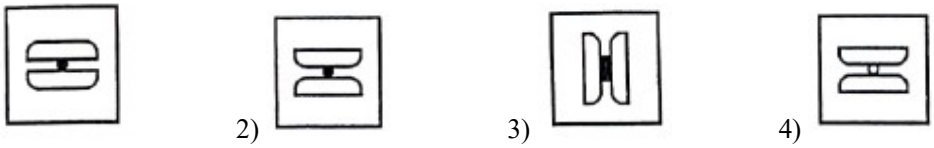

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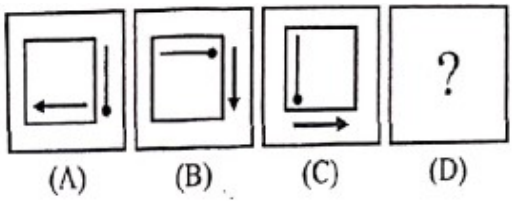

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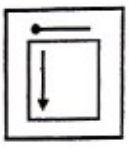
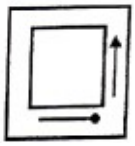
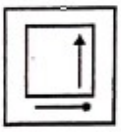
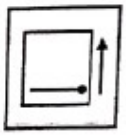
1)  2)  3)  4) 

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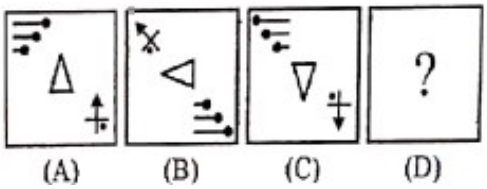

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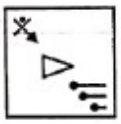
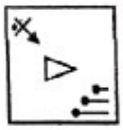
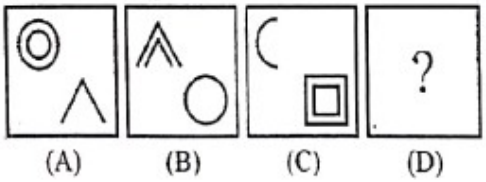
1)  2) 3)  4)

40. Problem Figures:



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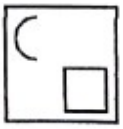
1)  2) 3) 4)

Direction: In Question nos. 41 to 50:

Questions have become wrong due to wrong order of signs. Choose the correct order of signs from the four alternatives given under each question, so that the equation becomes right. Write it in your answer sheet against the corresponding question number.

41. $30 \div 26 \times 8 = 7$

1) $\times - =$ 2) $- + =$ 3) $- \times =$ 4) $+ = \times$

42. $6 + 12 = 48 - 24$

1) $+ = \div$ 2) $\div = +$ 3) $+ - =$ 4) $\times - =$

43. $3 \times 8 = 19 \div 5$

1) $\times = +$ 2) $\div \times =$ 3) $\times + =$ 4) $= + -$

44. $17 = 3 + 43 \times 8$

1) $\times = +$ 2) $\div = +$ 3) $+ - =$ 4) $- = +$

45. $6 - 3 = 12 \div 6$

1) $= + -$ 2) $- = +$ 3) $\times = +$ 4) $+ = \div$

46. $5 = 4 + 11 - 9$

1) $- \times =$ 2) $+ = \div$ 3) $\times = +$ 4) $+ = -$

47. $48 + 8 \div 8 = 2$

1) $- \div =$ 2) $= + \times$ 3) $\div = -$ 4) $+ = \div$

48. $4 \times 9 = 6 + 1$

1) $\times - +$ 2) $= + -$ 3) $+ - =$ 4) $= - +$

49. $59 - 21 \times 8 = 10$

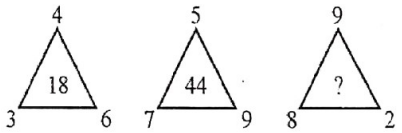
1) $+ = \times$ 2) $- = \times$ 3) $- = +$ 4) $= + -$

50. $22 = 14 + 48 - 12$

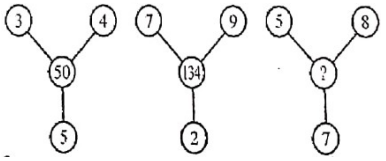
1) $\times = -$ 2) $= + \div$ 3) $- \times \div$ 4) $+ = -$

Direction: In question Nos. 51 to 60:

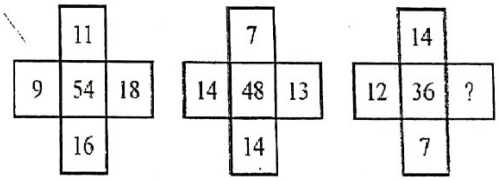
In these questions, numbers are arranged on the basis of some rules. One place is vacant, which is indicated as "?". Find out the correct alternatives to replace the question mark "?"

51. 

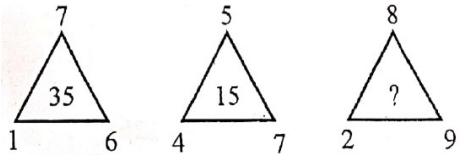
a) 77 b) 72 c) 56 d) 74

52. 

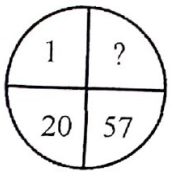
a) 140 b) 136 c) 144 d) 138

53. 

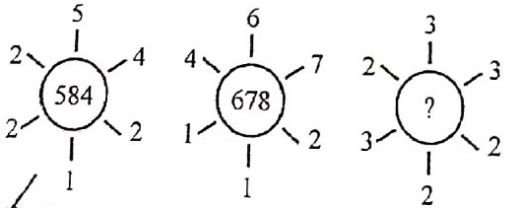
a) 11 b) 3 c) 4 d) 9

54. 

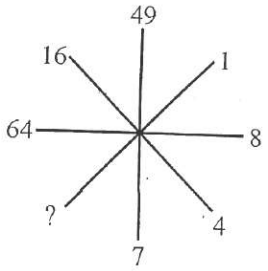
a) 56 b) 46 c) 74 d) 72

55. 

a) 115 b) 114 c) 121 d) 118

56. 

a) 694 b) 964 c) 649 d) 691

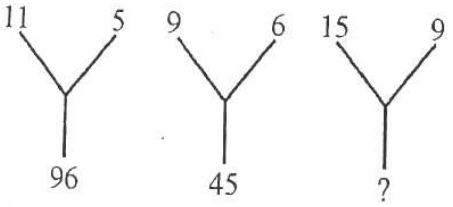
57. 

a) 2 b) 3 c) 4 d) 1

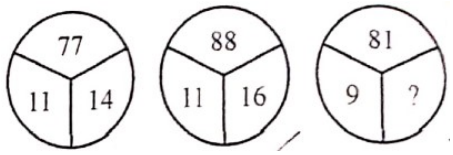
58.

3	2	1
6	5	4
7	8	9
39	38	?

a) 35 b) 36 c) 37 d) 41

59. 

a) 140 b) 135 c) 144 d) 138

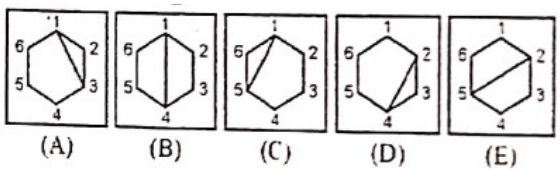
60. 

a) 27 b) 18 c) 16 d) 20

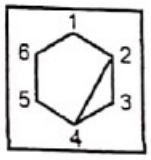
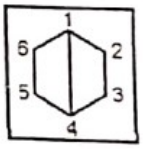
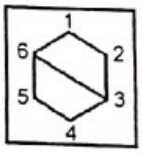
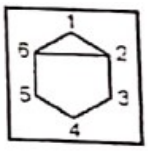
Direction: In question no.s 61 to 70

Each of the following questions consists of the five figures marked A, B, C, D and E called the problem figures followed by four alternatives marked 1, 2, 3 and 4 called the answer figures. Select a figure which will continue the same series established by the five problem figures.

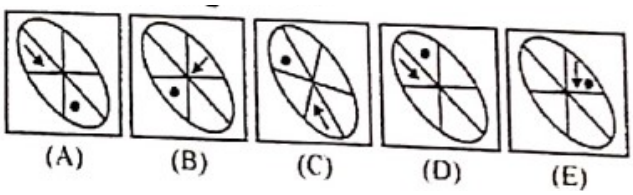
61. Problem figures



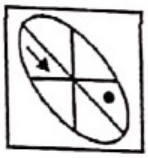
Answer figures

1)  2)  3)  4) 

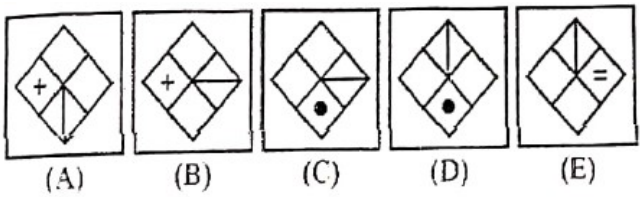
62. Problem figures



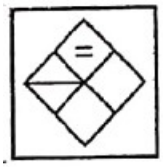
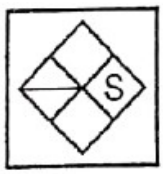
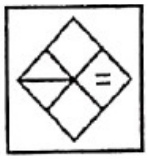
Answer figures

a)
b) 
c)
d)

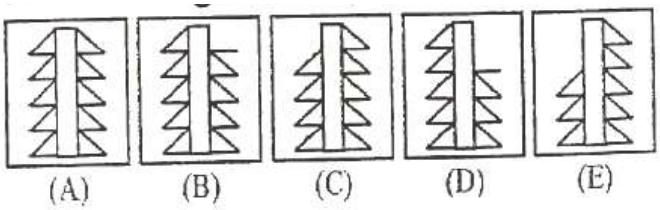
63. Problem figures



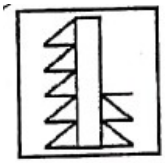
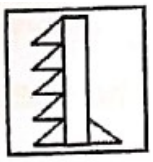
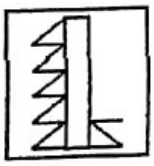
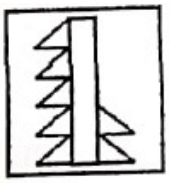
Answer figures

a)
b) 
c) 
d) 

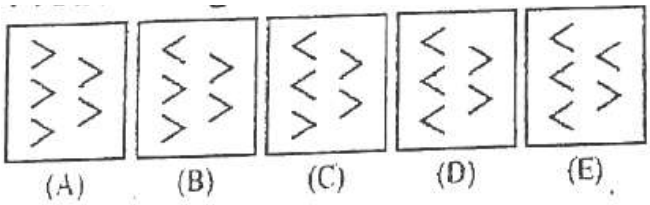
64. Problem figures



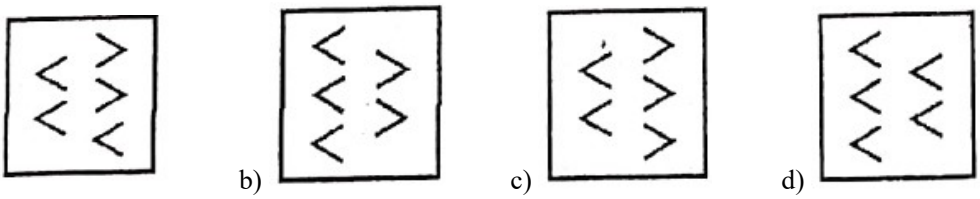
Answer figures

a) 
b) 
c) 
d) 

65. Problem figures

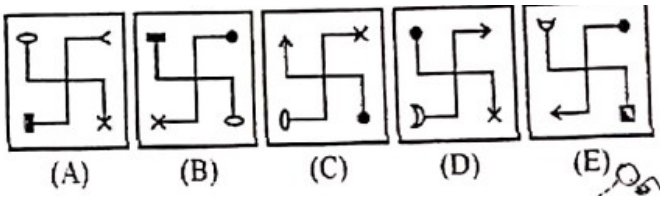


Answer figures

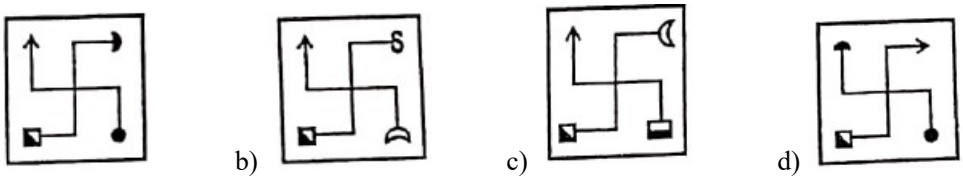


a) b) c) d)

66. Problem figures

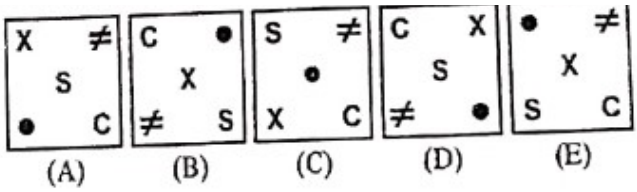


Answer figures

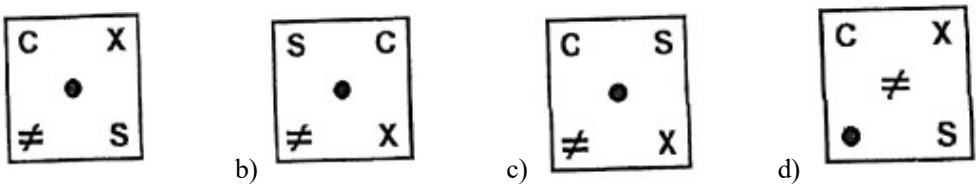


a) b) c) d)

67. Problem figures

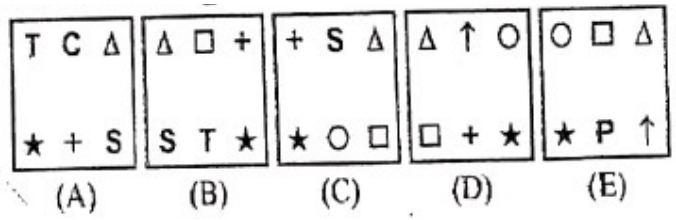


Answer figures

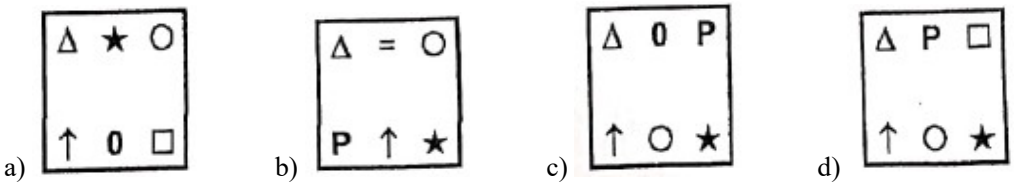


a) b) c) d)

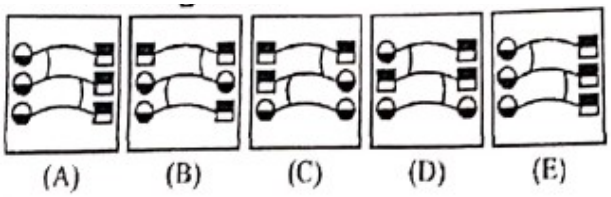
68. Problem figures



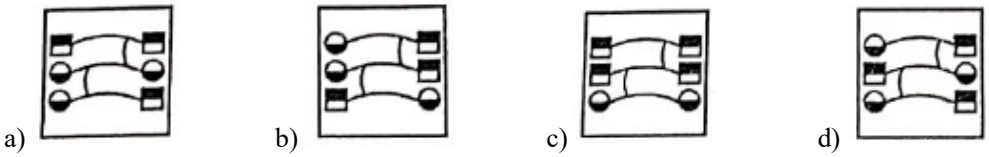
Answer figures



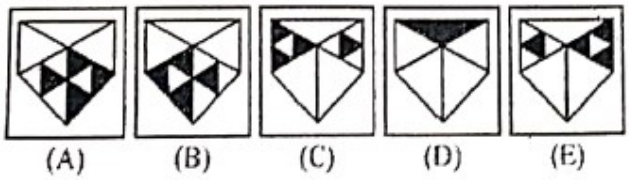
69. Problem figures



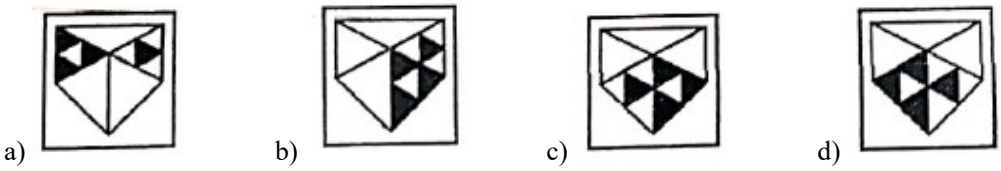
Answer figures



70. Problem figures



Answer figures



Directions: In Question nos. 71 to 75:

Read the following information carefully and answer the questions that follows:

(i) There is a group of five persons A, B, C, D and E

(ii) One of them is Teacher, one is a Doctor, one is a Journalist, one is an Industrialist and one is an advocate.

(iii) Three of them – A, C and advocate prefer tea to coffee and two of them – B and the journalist prefer coffee to tea.

(iv) The industrialist and D and A are friends to one another but two of these prefer coffee to tea.

(v) The teacher is C's brother.

71. Who is Teacher?
1) B 2) C 3) D 4) A

72. Which of the following groups includes a person who likes tea but is not an advocate?
1) B, C 2) B, D 3) A, B 4) A,C

73. Who is an Industrialist?
1) E 2) B 3) D 4) C

74. Who is a Doctor?
1) C 2) A 3) B 4) D

75. Which of the above statements is superfluous?
1) None 2) v 3) ii 4) iii

Directions: In question nos. 76 to 85:

There are four terms in each question. The term right to symbol : : have some relationship as the term of the left to the symbol : : and out of the four, one term is missing, which is among one of the given four alternatives, find the correct alternatives.

76. $AFC : DIF :: BED :$ _____
1) CGF 2) EIG 3) EHG 4) EGH

77. P.V. Sindhu : Badminton : : Deepak Puniya : _____
1) Wrestling 2) Boxing 3) Cricket 4) Hockey

78. Durand Cup : Foot ball : : Uber cup : _____
1) Badminton 2) Chess 3) Hockey 4) Cricket

79. PMG : SIX : : TIN : _____
1) WEE 2) BEE 3) WEB 4) SEE

80. Factory : Production : : School : _____
1) Building 2) Discipline 3) Teacher 4) Education

81. Earth : Moon : : _____ : Phobos
1) Venus 2) Mars 3) Jupiter 4) Mercury

82. Doctor : Patient : : Lawyer : _____
1) Client 2) Magistrate 3) Customer 4) Accused

83. Chandrayan : 2019 : : Mangalyan : _____
1) 2014 2) 2016 3) 2017 4) 2020

84. Gir : Gujarat : : Kaziranga : _____
1) Bengal 2) Kerala 3) Bihar 4) Assam

85. EART : CCPV : : BATH : _____
1) ZCRJ 2) DCBA 3) DZJR 4) CDBA

Direction: In question nos. 86 to 90:

Some letters are given in Column I and some digits are given in column II. Each digit of column II represents any letter of column I. Study the columns and write the alternative letter choosing the correct alternative against the corresponding question.

Column – I	Column – II
AMRVT	65479
RTHIB	46128
MRBZI	86317
IAMRV	85679
HIBZA	39218
MRTAB	47961
MVRTH	47526
RZIBH	38261
BZIRV	83156
IMVRH	75826

86. The sum of the codes H and Z is

1) 5 2) 10 3) 9 4) 6

87. The code for Z is

1) 2 2) 8 3) 3 4) 1

88. The code for I is

1) 2 2) 7 3) 8 4) 1

89. The code for H is

1) 3 2) 6 3) 2 4) 1

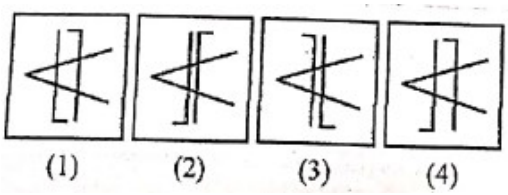
90. The code for A is

1) 6 2) 3 3) 9 4) 8

Direction: In question nos. 91 to 100:

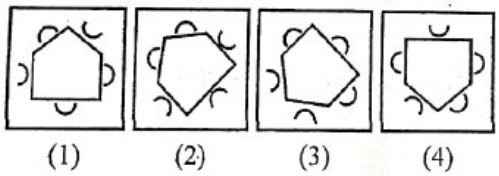
Out of the four figures (1) , (2), (3), (4) given in each question, three are similar in a certain way. Choose the figure which is different from the other figures.

91. Choose the figure which is different from the rest.



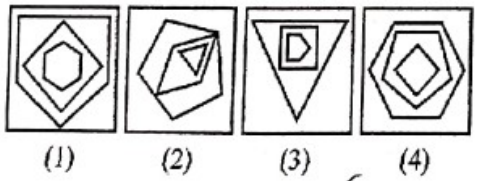
1) 2 2) 3 3) 4 4) 1

92. Choose the figure which is different from the rest.



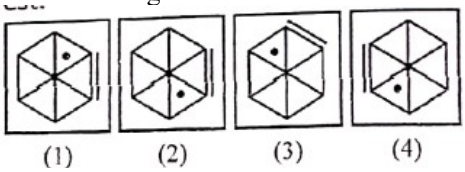
1) 4 2) 2 3) 3 4) 1

93. Choose the figure which is different from the rest.



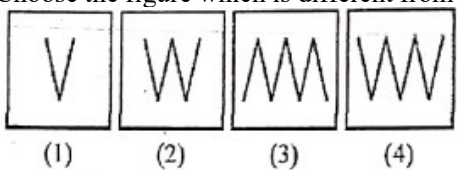
1) 2 2) 1 3) 4 4) 3

94. Choose the figure which is different from the rest:



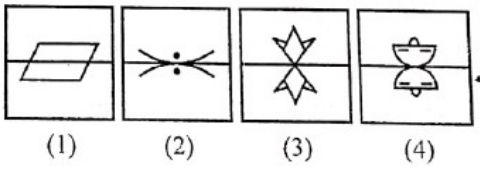
1) 2 2) 4 3) 3 4) 1

95. Choose the figure which is different from the rest.



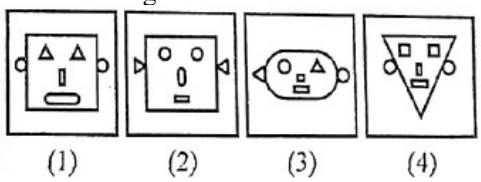
1) 4 2) 2 3) 3 4) 1

96. Choose the figure which is different from the rest.



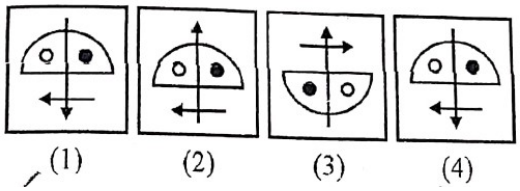
1) 2 2) 1 3) 4 4) 3

97. Choose the figure which is different from the rest.



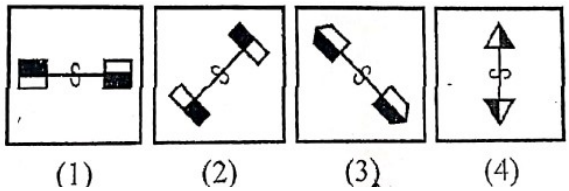
1) 2 2) 1 3) 3 4) 4

98. Choose the figure which is different from the rest.



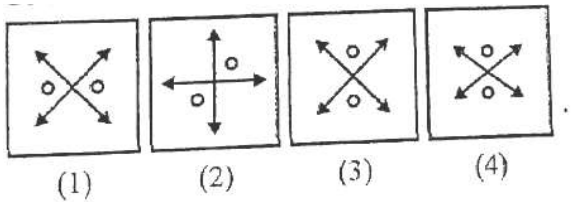
1) 2 2) 1 3) 3 4) 4

99. Choose the figure which is different from the rest.



1) 3 2) 2 3) 4 4) 1

100. Choose the figure which is different from the rest.



1) 3 2) 1 3) 4 4) 2